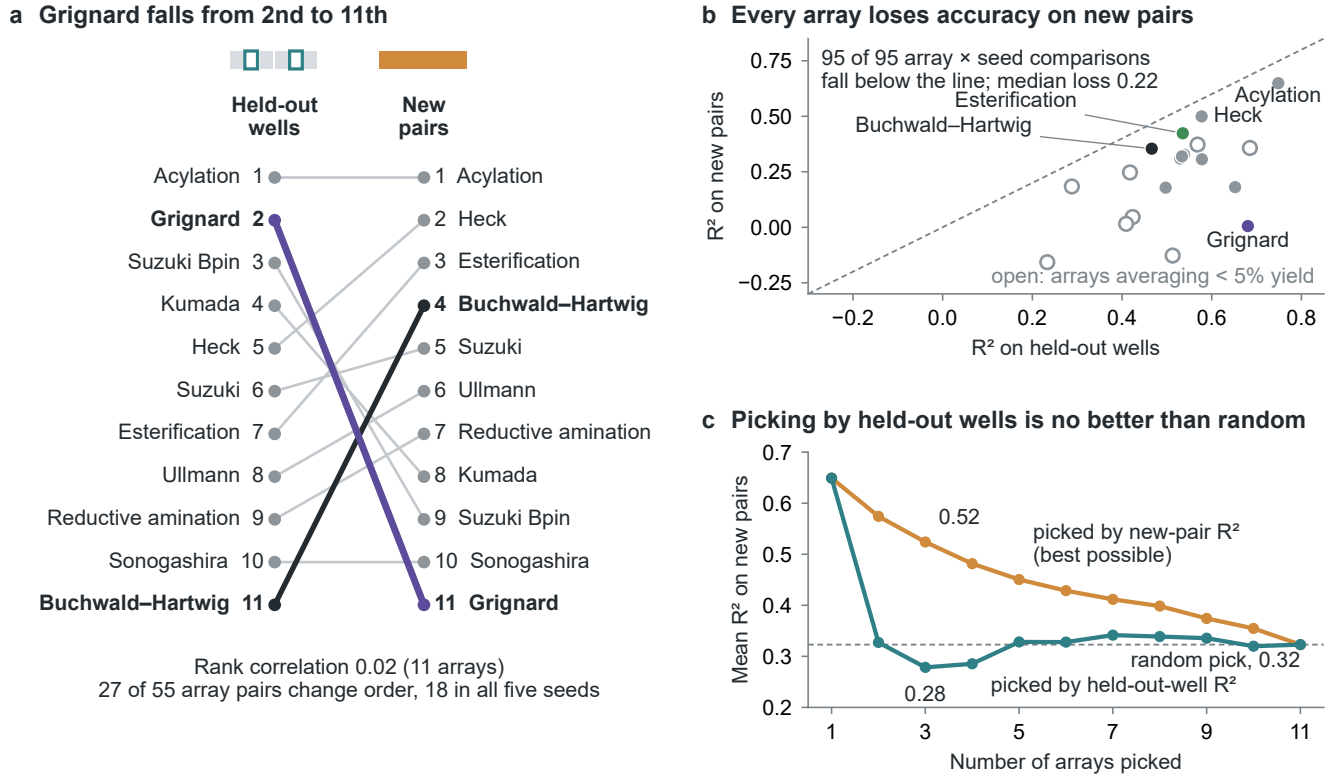

a Grignard falls from 2nd to 11th
Held-out wells
New pairs
Acylation 1
Grignard 2
Suzuki Bpin 3
Kumada 4
Heck 5
Suzuki 6
Esterification 7
Ullmann 8
Reductive amination 9
Sonogashira 10
Buchwald–Hartwig 11
1 Acylation
2 Heck
3 Esterification
4 Buchwald–Hartwig
5 Suzuki
6 Ullmann
7 Reductive amination
8 Kumada
9 Suzuki Bpin
10 Sonogashira
11 Grignard
Rank correlation 0.02 (11 arrays)
27 of 55 array pairs change order, 18 in all five seeds
b Every array loses accuracy on new pairs
95 of 95 array × seed comparisons fall below the line; median loss 0.22
Esterification
Buchwald–Hartwig
Acylation
Heck
Grignard
open: arrays averaging < 5% yield
R² on new pairs
R² on held-out wells
0.75
0.50
0.25
0.00
−0.25
−0.2
0.0
0.2
0.4
0.6
0.8
c Picking by held-out wells is no better than random
Mean R² on new pairs
Number of arrays picked
0.52
picked by new-pair R² (best possible)
random pick, 0.32
0.28
picked by held-out-well R²
0.7
0.6
0.5
0.4
0.3
0.2
1
3
5
7
9
11